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(Updated on October 2026)

Brief Bio

Aykut Erdem is an Associate Professor of Computer Engineering at Koç University, affiliated with the KUIS AI Center, and a member of ELLIS (European Laboratory for Learning and Intelligent Systems). Prior to joining Koç University, he was a faculty member in the Department of Computer Engineering at Hacettepe University, where he co-directed the Computer Vision Lab. His research focuses on generative and multimodal artificial intelligence, with particular interests in visual generation and understanding, vision-language learning, and multimodal reasoning. He received the Science Academy's Young Scientist Award (BAGEP) in Computer Engineering in 2021 and the Koç University Outstanding Faculty Award in 2025. Since 2022, he has served on the editorial board of the IEEE Transactions on Image Processing, where he currently serves as a Senior Area Editor.

Professional Experience

- **Associate Professor**^{1,2}, 09/2020 – now
Department of Computer Engineering, School of Engineering, Koç University, Sariyer, Istanbul, Turkey
- **Associate Professor**^{3,4} 04/2018 – 08/2020
Department of Computer Engineering, Hacettepe University, Beytepe, Ankara, Turkey
- **Assistant Professor** 12/2011 – 04/2018
Department of Computer Engineering, Hacettepe University, Beytepe, Ankara, Turkey
- **Instructor** 11/2010 – 12/2011
Department of Geodesy and Photogrammetry, Hacettepe University, Beytepe, Ankara, Turkey
- **Post-doctoral Researcher**⁵ 11/2008 – 03/2010
Dipartimento di Informatica, Università Ca' Foscari di Venezia, Venice, Italy

PhD in Computer Engineering

09/2003–10/2008

METU, Ankara, Turkey

MSc in Computer Engineering

09/2001–07/2003

METU, Ankara, Turkey

BSc in Computer Engineering

09/1997–06/2001

METU, Ankara, Turkey

¹ Affiliated faculty at Koç University and İş Bankası Artificial Intelligence Laboratory (KUIS AI Center).

² Taught courses on *computer systems and programming, deep learning, deep unsupervised learning, and computational imaging*.

³ A co-founder of Hacettepe University Computer Vision Lab (HUCVL).

⁴ Taught courses on *introductory programming, logical design, programming languages, systems programming, machine learning, image processing, computer vision, machine learning learning and deep learning*.

⁵ Sponsored by EU-FP7 SIMBAD (Beyond Features: Similarity-Based Pattern Analysis and Recognition) Project.

- **Visiting Student**⁶, 09/2007 – 11/2007
Department of Architecture, Computation Group, MIT, Cambridge, USA
 - **Visiting Student**⁷, 07/2006
Department of Mathematics, Northeastern University, Boston, USA
 - **Visiting Research Scholar**⁸, 07/2004 – 08/2004
Virginia Bioinformatics Institute, Virginia Polytechnic Institute and State University, Virginia, USA
 - **Teaching Assistant**⁹, 12/2001 – 09/2008
Department of Computer Engineering, Middle East Technical University, Ankara, Turkey
- ⁶ Visited Prof. George Stiny (sponsored by TUBITAK Program 2214 – Research Fellowship Program for PhD Students).
- ⁷ Visited Prof. Jayant Shah.
- ⁸ Worked with Volkan Atalay and Rengul Cetin-Atalay.
- ⁹ Assisted courses include *image processing, dynamic systems, scientific computing, logic for CS, and computer graphics*.

Publications

Journal Articles

My Erdos number is 3 (via 1. Arun Kumar Jagota, 2. Marcello Pelillo).

1. Onur Eker, Erkut Erdem, Aykut Erdem, “EeveeDark: A Binary Neural Framework for Low-Light Video Enhancement via Event-Guided Sensor-Level Fusion”, *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4633-4640, April 2026.
2. Mahmut Esat Demirhan, Seniha Esen Yuksel, Senior Member, Erkut Erdem, Aykut Erdem, Anna-Maria Raita-Hakola, and Ilkka Pölönen, “Self-Supervised Low-Light Hyperspectral Image Enhancement via Fourier-Based Transformer Network”, *IEEE Journal of Selected Topics in Signal Processing*, doi: 10.1109/JSTSP.2025.3632537, November 2025.
3. Mert Cokelek, Halit Ozsoy, Nevrez Imamoglu, Cagri Ozcinar, Inci Ayhan, Erkut Erdem, Aykut Erdem, “Spherical Vision Transformers for Audio-Visual Saliency Prediction in 360° Videos”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 48, Issue 1, pp. 329 - 345, January 2026.
4. Nafiseh Jabbari Tofighi, M. Hedi Elfkir, Nevrez Imamoglu, Cagri Ozcinar, Aykut Erdem, Erkut Erdem, “Omnidirectional Image Quality Assessment with Local-Global Vision Transformers”, *Image and Vision Computing*, Vol. 148, 105151, August 2024.
5. Elena Lloret, Anabela Barreiro, Mehul Bhatt, Alberto Bugarín-Diz, Gianfranco E. Modoni, Max Silberztein, Iacer Calixto, Grazina Korvel, Konstantinos Diamantaras, Alkiviadis Katsalis, Oleksii Turuta, Irene Russo, Aykut Erdem, “Multi3Generation: Multitask, Multilingual, and Multimodal Language Generation”, *Open Research Europe* 2024, 3:176
6. Ahmet E. Tekden, Aykut Erdem, Erkut Erdem, Tamim Asfour, Emre Ugur, “Object and relation centric representations for push effect prediction”, *Robotics and Autonomous Systems*, Vol. 174, April 2024.
7. Burak Ercan, Onur Eker, Canberk Saglam, Aykut Erdem, Erkut Erdem, “HyperE2VID: Improving Event-Based Video Reconstruction via Hypernetworks”, *IEEE Transactions on Image Processing*, Vol. 33, pp. 1826-1837, March 2024.
8. Orhan Torun, Seniha Esen Yuksel, Erkut Erdem, Nevrez Imamoglu, Aykut Erdem, “Hyperspectral Image Denoising via Self-Modulating Convolutional Neural Networks”, *Signal Processing*, Vol. 214, January 2024.

9. Ahmet Canberk Baykal, Abdul Basit Anees, Duygu Ceylan, Erkut Erdem, Aykut Erdem, Deniz Yuret, "CLIP-guided StyleGAN Inversion for Text-driven Real Image Editing", *ACM Transactions on Graphics*, Vol. 42, Issue 5, Article 172, August 2023.
10. Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, et al., "Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models", *Transactions on Machine Learning Research*, May 2023.
11. Aysun Kocak, Erkut Erdem, Aykut Erdem, "A Gated Fusion Network for Dynamic Saliency Prediction", *IEEE Transactions on Cognitive and Developmental Systems*, Vol. 14, Issue 3, pp. 995-1008, September 2022.
12. Erkut Erdem, Menekse Kuyu, Semih Yagcioglu, Anette Frank, Letitia Parcalabescu, Andrii Babii, Oleksii Turuta, Aykut Erdem, Iacer Calixto, Barbara Plank, Elena Lloret, Elena-Simona Apostol, Ciprian-Octavian Truică, Branislava Šandrih, Sanda Martinčić-Ipšić, Gábor Berend, Albert Gatt, Grazina Korvel, "Neural Natural Language Generation: A Survey on Multilinguality, Multimodality, Controllability and Learning", *Journal of Artificial Intelligence Research*, Vol. 73, pp. 1131-1207, April 2022.
13. Kemal Cizmeciler, Erkut Erdem, Aykut Erdem, "Leveraging semantic saliency maps for query-specific video summarization", *Multimedia Tools and Applications*, March 2022.
14. Ahmet Serdar Karadeniz, Erkut Erdem, Aykut Erdem, "Burst Photography for Learning to Enhance Extremely Dark Images", *IEEE Transactions on Image Processing*, Vol. 30, pg. 9372-9385, November 2021.
15. Murat Ture, M. Ege Ciklabakkal, Aykut Erdem, Erkut Erdem, Pinar Satilmis, Ahmet Oguz Akyuz, "From Noon to Sunset: Interactive Rendering, Relighting, and Recoloring of Landscape Photographs by Modifying Solar Position", *Computer Graphics Forum*, Vol. 40, Issue 6, pp. 500-515, September 2021.
16. Onur Can Uner, Cem Aslan, Burak Ercan, Tayfun Ates, Ufuk Celikcan, Aykut Erdem, Erkut Erdem, "Synthetic18K: Learning better representations for person re-ID and attribute recognition from 1.4 million synthetic images", *Signal Processing: Image Communication*, Vol. 97, September 2021.
17. Abdulrahman Haje Kerim, Cem Aslan, Ufuk Celikcan, Erkut Erdem, Aykut Erdem, "NOVA: Rendering Virtual Worlds with Humans for Computer Vision Tasks", *Computer Graphics Forum*, Vol. 40, Issue 6, pp. 258-272, September 2021.
18. Abdulrahman Kerim, Ufuk Celikcan, Erkut Erdem, Aykut Erdem, "Using Synthetic Data for Person Tracking Under Adverse Weather Conditions", *Image and Vision Computing*, Vol. 111, July 2021.
19. Begum Citamak, Ozan Caglayan, Menekse Kuyu, Erkut Erdem, Aykut Erdem, Pranava Madhyastha, and Lucia Specia, "MSVD-Turkish: A Comprehensive Multimodal Video Dataset for Integrated Vision and Language Research in Turkish", *Machine Translation*, 35(2), 265-288, June 2021.
20. Emre Boran, Aykut Erdem, Nazli Ikizler-Cinbis, Erkut Erdem, Pranava Madhyastha, Lucia Specia, "Leveraging auxiliary image descriptions for dense video captioning", *Pattern Recognition Letters*, Vol. 146, June 2021.
21. Mahmut Yurt, Salman UH Dar, Aykut Erdem, Erkut Erdem, Kader K. Oguz, Tolga Cukur, "mustGAN: multi-stream Generative Adversarial Networks for MR Image Synthesis", *Medical Image Analysis*, Vol. 70, May 2021.
22. Bora Celikkale, Goksu Erdogan, Aykut Erdem, and Erkut Erdem, "Generating Visual Story Graphs with Application to Photo Album Summarization", *Signal Processing: Image Communication*, Vol. 90, January 2021.

23. Yasin Kavak, Erkut Erdem, and Aykut Erdem, "Hedging Static Saliency Models to Predict Dynamic Saliency", *Signal Processing: Image Communication*, Vol. 81, February 2020.
24. Levent Karacan, Zeynep Akata, Aykut Erdem, and Erkut Erdem, "Manipulating Attributes of Natural Scenes via Hallucination", *ACM Transactions on Graphics*, Vol. 39, Issue 1, Article 7, February 2020.
25. Salman Ul Hassan Dar, Mahmut Yurt, Levent Karacan, Aykut Erdem, Erkut Erdem, Tolga Cukur, "Image Synthesis in Multi-Contrast MRI with Conditional Generative Adversarial Networks", *IEEE Transactions on Medical Imaging*, Vol. 38, Issue 10, pp. 2375-2388, October 2019.
26. Cagdas Bak, Aysun Kocak, Erkut Erdem, and Aykut Erdem, "Spatio-Temporal Saliency Networks for Dynamic Saliency Prediction", *IEEE Transactions on Multimedia*, Vol. 20, Issue 7, pp. 1688-1698, July 2018.
27. Levent Karacan, Aykut Erdem, and Erkut Erdem, "Alpha Matting with KL-Divergence Based Sparse Sampling", *IEEE Transactions on Image Processing*, Vol. 26, Issue 9, pp. 4523-4536, September 2017.
28. Mert Kilickaya, Burak Kerim Akkus, Ruket Cakici, Aykut Erdem, Erkut Erdem, and Nazli Ikizler-Cinbis, "Data-driven image captioning via salient region discovery", *IET Computer Vision*, Vol. 11, Issue 6, pp. 398-406, September 2017.
29. Yasin Kavak, Erkut Erdem, and Aykut Erdem, "A Comparative Study for Feature Integration Strategies in Dynamic Saliency Estimation", *Signal Processing: Image Communication*, Vol. 51, pp. 13-25, February 2017.
30. Aykut Erdem, "Structure-Texture Decomposition of RGB-D Images", *International Journal of Intelligent Systems and Applications in Engineering*, Vol. 4, No. 4, pp. 111-118, October 2016.
31. Osman Akin, Erkut Erdem, Aykut Erdem, and Krystian Mikolajczyk, "Deformable Part-based Tracking by Coupled Global and Local Correlation Filters", *Journal of Visual Communication and Image Representation*, Vol. 38, pp. 763-774, July 2016.
32. Okan Tarhan Tursun, Ahmet Oguz Akyuz, Aykut Erdem, and Erkut Erdem, "An Objective Deghosting Quality Metric for HDR Images", *Computer Graphics Forum*, Vol. 35, Issue 2, pp. 139-152, May 2016.
33. Raffaella Bernardi, Ruket Cakici, Desmond Elliott, Aykut Erdem, Erkut Erdem, Nazli Ikizler-Cinbis, Frank Keller, Adrian Muscat, and Barbara Plank, "Automatic Description Generation from Images: A Survey of Models, Datasets, and Evaluation Measures", *Journal of Artificial Intelligence Research*, Vol. 55, pp. 409-442, February 2016.
34. Bora Celikkale, Aykut Erdem and Erkut Erdem, "Predicting Memorability of Images Using Attention-driven Spatial Pooling and Image Semantics", *Image and Vision Computing*, Vol. 42, pp. 35-46, October 2015.
35. Okan Tarhan Tursun, Ahmet Oguz Akyuz, Aykut Erdem and Erkut Erdem, "The State of the Art in HDR Deghosting: A Survey and Evaluation", *Computer Graphics Forum*, Vol. 34, Issue 2, pp. 683-707, May 2015.
36. Levent Karacan, Erkut Erdem and Aykut Erdem, "Structure-Preserving Image Smoothing via Region Covariances", *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia 2013)*, Vol. 32, Issue 6, November 2013.
37. Erkut Erdem and Aykut Erdem, "Visual saliency estimation by nonlinearly integrating features using region covariances", *Journal of Vision*, Vol. 13, Issue 4, Article 11, March 2013.
38. Aykut Erdem and Marcello Pelillo, "Graph Transduction as a Non-Cooperative Game", *Neural Computation*, Vol. 24, No. 3, pp. 700-723, March 2012.

39. Aykut Erdem and Sibel Tari, "A similarity-based approach for shape classification using Aslan skeletons", *Pattern Recognition Letters*, Vol. 31, Issue 13, pp. 2024-2032, October 2010.
40. Emre Baseski, Aykut Erdem and Sibel Tari, "Dissimilarity Between Two Skeletal Trees in a Context", *Pattern Recognition*, Vol. 42, Issue 3, pp. 370-385, March 2009.
41. Cagri Aslan, Aykut Erdem, Erkut Erdem and Sibel Tari, "Disconnected Skeleton: Shape at its Absolute Scale", *IEEE Trans. Pattern Anal. Mach. Intell.*, Vol. 30, No. 12, pp. 2188-2203, December 2008.
42. Aykut Erdem, Erkut Erdem, Volkan Atalay and A. Enis Cetin, "Vision-based continuous Graffiti-like text entry system", *Optical Engineering*, Vol. 43, Issue 3, pp. 553-558, March 2004.

Conference/Workshop Publications

1. Muhammed Burak Kızıl, Enes Sanli, Niloy Mitra, Xuelin Chen, Erkut Erdem, Aykut Erdem, Duygu Ceylan. "Auteur: Language-Driven Cinematographic Framing for Human-Centric Video Generation", *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, Sydney, Australia, December 2026.
2. Andrew Bond, Ege Erdem Özlü, Tuna Çimen, Ilkin Umut Melanlıoglu, Tolga Birdal, Erkut Erdem, Aykut Erdem. "Task-Induced Riemannian Metrics for Vision Transformer Feature Spaces", *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, Sydney, Australia, December 2026.
3. Yigit Ekin, Enes Sanli, Aykut Erdem, Erkut Erdem, and Aysegul Dundar. "BeyondMasks: Evaluating Causal and Physical Consistency in Video Object Removal", *The Nineteenth European Conference on Computer Vision (ECCV 2026)*, Malmo, Sweden, September 2026.
4. Ezel Bayraktar, Alperen Demirci, Erkut Erdem, Aykut Erdem, Mukhamediyar Amanzhol, Adam Sattout, "CityViews: Benchmarking Satellite and Street-Level Imagery in Urban Vision-Language Models", *The second Workshop on Geospatial AI and Foundation Models (GAIA 2026)* – in conjunction with ECCV 2026, Malmo, Sweden, September 2026.
5. M. Burak Kizil, Enes Sanli, Niloy J. Mitra, Erkut Erdem, Aykut Erdem, Duygu Ceylan, "LAMP: Language-Assisted Motion Planning for Controllable Video Generation", *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)*, Denver, Colorado, USA, June 2026.
6. Alperen Demirci, Ezel Bayraktar, Bora Dere, Erkut Erdem, Aykut Erdem, "When the Right Answer Is Missing: Probing Hallucination in Medical Reasoning Models via NOTA Evaluation", *CVPR 2026 Workshop on Medical Reasoning with Vision Language Foundation Models (Med-Reasoner)* – conjunction with CVPR 2026, June 2026.
7. Yakup Abrek Er, Ilker Kesen, Gözde Gül Şahin, Aykut Erdem, "Cetvel: A Unified Benchmark for Evaluating Language Understanding, Generation and Cultural Capacity of LLMs for Turkish", *The 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*, Rabat, Morocco March 2026.
8. Camila Zurdo Tagliabue, Heloisa Oss Boll, Aykut Erdem, Erkut Erdem, Iacer Calixto, "DeVisE: Towards the Behavioral Testing of Medical Large Language Models", *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*, Rabat, Morocco March 2026.
9. Ali Kerem Bozkurt, Baris Cem Bakay, Ibrahim Kulac, Cigdem Gunduz-Demir, Erkut Erdem, Aykut Erdem, "Scalable Whole-Slide Vision-Language Modeling with Learned Token Pruning", *The Second*

- Workshop on GenAI for Health: Potential, Trust and Policy Compliance at the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*, Vancouver, December 2025.
10. Andrew Bond, Jui-Hsien Wang, Long Mai, Erkut Erdem, Aykut Erdem, "GaussianVideo: Efficient Video Representation via Hierarchical Gaussian Splatting", *IEEE International Conference on Computer Vision (ICCV 2025)*, Honolulu, Hawaii, October 2025.
 11. Enes Karanfil, Nevrez Imamoglu, Erkut Erdem, Aykut Erdem, "A Vision-Language Framework for Multispectral Scene Representation Using Language-Grounded Features", *2025 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2025)*, Brisbane, Australia, August 2025.
 12. Tolga Ozdemir, Erkut Erdem, Orhan Torun, Seniha Esen Yuksel, Aykut Erdem, "HSDSSM: A Hybrid Spectral Denoising State-Space Model for Hyperspectral Images", *2025 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2025)*, Brisbane, Australia, August 2025.
 13. Damla Akcaoglu, Erkut Erdem, Seniha Esen Yuksel, Aykut Erdem, "Enhancing Hyperspectral Image Synthesis via the Spectral Super-resolution Post-Processing Network", *2025 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2025)*, Brisbane, Australia, August 2025.
 14. Emre Can Acikgoz, Osman Batur Ince, Rayene Bech, Arda Anil Boz, Ilker Kesen, Aykut Erdem, Erkut Erdem, "Hippocrates: An Open-Source Framework for Advancing Large Language Models in Healthcare", *Workshop on GenAI for Health: Potential, Trust and Policy Compliance at the Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*, Vancouver, December 2024.
 15. Yigit Ekin, Ahmet Burak Yildirim, Erdem Eren Caglar, Aykut Erdem, Erkut Erdem, Aysegul Dundar, "CLIPAway: Harmonizing Focused Embeddings for Removing Objects via Diffusion Models", *the Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*, Vancouver, Canada, Vancouver, December 2024.
 16. Abdul Basit Anees, Ahmet Canberk Baykal, Muhammed Burak Kizil, Duygu Ceylan, Erkut Erdem, Aykut Erdem, "HyperGAN-CLIP: A Unified Framework for Domain Adaptation, Image Synthesis and Manipulation", *SIGGRAPH Asia 2024 Conference Papers*, Tokyo, Japan, December 2024.
 17. Burak Ercan, Onur Eker, Aykut Erdem, and Erkut Erdem, "HUE Dataset: High-Resolution Event and Frame Sequences for Low-Light Vision", *Workshop on Neuromorphic Vision (NeVi): Advantages and Applications of Event Cameras at ECCV 2024*, Milan, Italy, pp. 174-191, September 2024.
 18. Orhan Torun, S. Esen Yuksel, Erkut Erdem, Aykut Erdem, "Self-Supervised Calibration of the Denoising Networks for HSI", *2024 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2024)*, Athens, Greece, July 2024.
 19. Semih Yagcioglu, Osman Batur Ince, Aykut Erdem, Erkut Erdem, Desmond Elliott, Deniz Yuret, "Compositional Generalization in Multimodal Models", *2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024)*, June 2024.
 20. Ilker Kesen, Andrea Pedrotti, Mustafa Dogan, Michele Cafagna, Emre Can Acikgoz, Letitia Parcalabescu, Iacer Calixto, Anette Frank, Albert Gatt, Aykut Erdem, Erkut Erdem, "ViLMA: A Zero-Shot Benchmark for Linguistic and Temporal Grounding in Video-Language Models", *The Twelfth International Conference on Learning Representations (ICLR 2024)*, Vienna, May 2024.
 21. Ahmet Onur Dasdemir, Can Dimici, Aykut Erdem, and Emir Salih Magden, "Physically Accurate Fast Nanophotonic Simulations with Physics Informed Model and Training", *Workshop on Machine Learning and the Physical Sciences at the 37th Conference on Neural Information Processing Systems (NeurIPS)*, December 2023.

22. Osman İnce, Tanin Zeraati, Semih Yagcioglu, Yadollah Yaghoobzadeh, Erkut Erdem, and Aykut Erdem, Harnessing Dataset Cartography for Improved Compositional Generalization in Transformers, *Findings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, Singapore, December 2023.
23. Mert Cokelek, Nevrez Imamoglu, Cagri Ozcinar, Erkut Erdem, Aykut Erdem, “Spherical Vision Transformer for 360-degree Video Saliency Prediction”, *The 34th British Machine Vision Conference (BMVC)*, Aberdeen, UK, November 2023.
24. Moayed Haji Ali, Andrew Bond, Tolga Birdal, Duygu Ceylan, Levent Karacan, Erkut Erdem, Aykut Erdem, “VidStyleODE: Disentangled Video Editing via StyleGAN and NeuralODEs”, *IEEE International Conference on Computer Vision (ICCV 2023)*, Paris, October 2023.
25. Burak Ercan, Onur Eker, Aykut Erdem, Erkut Erdem, “EVREAL: Towards a Comprehensive Benchmark and Analysis Suite for Event-based Video Reconstruction”, *4th International Workshop on Event-Based Vision at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)*, Vancouver, June 2023.
26. Nafiseh Tofighi, M. Hedi Elfkir, Nevrez Imamoglu, Cagri Ozcinar, Erkut Erdem, Aykut Erdem, “ST360IQ: No-Reference Omnidirectional Image Quality Assessment with Spherical Vision Transformers”, *2023 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2023)*, Greece, June 2023.
27. Ilker Kesen, Aykut Erdem, Erkut Erdem and Iacer Calixto, “Detecting Euphemisms with Literal Descriptions and Visual Imagery”, *The Third Workshop on Figurative Language Processing (FigLang 2022)* – in conjunction with EMNLP 2022, Abu Dhabi, December 2022.
28. Levent Karacan, Tolga Kerimoğlu, I. Ata Inan, Tolga Birdal, Erkut Erdem, Aykut Erdem, “Disentangling Content and Motion for Text-Based Neural Video Manipulation”, *The 33rd British Machine Vision Conference (BMVC)*, London, UK, November 2022.
29. Cansu Korkmaz, A. Murat Tekalp, Zafer Doğan, Erkut Erdem, Aykut Erdem, “Perception-Distortion Tradeoff in the SR Space Spanned by Flow Models”, *The 29th IEEE International Conference on Image Processing (IEEE ICIP)*, Bordeaux, France, October 2022.
30. Ilker Kesen, Ozan Arkan Can, Erkut Erdem, Aykut Erdem, Deniz Yuret, “Modulating Bottom-Up and Top-Down Visual Processing via Language-Conditional Filters”, *5th Multimodal Learning and Applications Workshop (MULA 2022)* – in conjunction with CVPR 2022 (**Best Paper Award**), New Orleans, USA, June 2022.
31. Anabela Barreiro, José GC de Souza, Albert Gatt, Mehul Bhatt, Elena Lloret, Aykut Erdem, Dimitra Gkatzia, Helena Moniz, Irene Russo, Fabio Kepler, Iacer Calixto, Marcin Paprzycki, François Portet, Isabelle Augenstein, Mirela Alhasani, “Multi3Generation: Multitask, Multilingual, Multimodal Language Generation”, *23rd Annual Conference of the European Association for Machine Translation (EAMT 2022)*, Ghent, Belgium, June 2022.
32. Tayfun Ates, Muhammed Samil Atesoglu, Cagatay Yigit, Ilker Kesen, Mert Kobas, Erkut Erdem, Aykut Erdem, Tilbe Goksun, Deniz Yuret, “CRAFT: A Benchmark for Causal Reasoning About Forces and inTeractions”, *Findings of the Association for Computational Linguistics (ACL 2022)*, May 2022.
33. Adil Kaan Akan, Erkut Erdem, Aykut Erdem, Fatma Guney, “SLAMP: Stochastic Latent Appearance and Motion Prediction”, *IEEE International Conference on Computer Vision (ICCV 2021)*, Virtual, October 2021.

34. Mert Cokelek, Nevrez Imamoglu, Cagri Ozcinar, Erkut Erdem, Aykut Erdem, "Leveraging Frequency Based Salient Spatial Sound Localization to Improve 360° Video Saliency Prediction", *The 17th International Conference on Machine Vision Applications (MVA 2021)*, Nagoya, Japan, July 2021.
35. Ozan Caglayan, Menekse Kuyu, Mustafa Sercan Amac, Pranava Madhyastha, Erkut Erdem, Aykut Erdem, Lucia Specia, "Cross-lingual Visual Pre-training for Multimodal Machine Translation", *The 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2021)*, Kyiv, Ukraine, April 2021.
36. Tayfun Ates, Muhammed Samil Atesoglu, Cagatay Yigit, Ilker Kesen, Mert Kobas, Erkut Erdem, Aykut Erdem, Tilbe Goksun, Deniz Yuret, "CRAFT: A Benchmark for Causal Reasoning About Forces and Interactions", *2nd Workshop on Shared Visual Representations in Human and Machine Intelligence (SVRHM) at the 34th Conference on Neural Information Processing Systems (NeurIPS 2020)*, December 2020.
37. Ahmet E. Tekden, Aykut Erdem, Erkut Erdem, Mert Imre, M. Yunus Seker, Emre Ugur, "Belief Regulated Dual Propagation Nets for Learning Action Effects on Articulated Multi-Part Objects", *International Conference on Robotics and Automation (ICRA 2020)*, Paris, France, May 2020.
38. Levent Karacan, Mehmet Gunel, Aykut Erdem, Erkut Erdem, "Manipulating Person Videos with Natural Language", *Sets & Partitions Workshop at the 33rd Conference on Neural Information Processing Systems (NeurIPS 2019)*, Vancouver, Canada, December 2019.
39. Mustafa Sercan Amac, Semih Yagcioglu, Aykut Erdem, and Erkut Erdem, "Procedural Reasoning Networks for Understanding Multimodal Procedures", *The SIGNLL Conference on Computational Natural Language Learning (CoNLL 2019)*, Hong Kong, November 2019.
40. Yunus Emre Ozkose, Bora Celikkale, Aykut Erdem, and Erkut Erdem, "Diverse Neural Photo Album Summarization", *International Conference on Image Processing Theory, Tools and Applications (IPTA)*, Istanbul, November 2019.
41. Semih Yagcioglu, Aykut Erdem, Erkut Erdem, and Nazli Ikizler-Cinbis, "RecipeQA: A Challenge Dataset for Multimodal Comprehension of Cooking Recipes", *2018 Conference on Empirical Methods in Natural Language Processing (EMNLP 2018)*, Brussels, Belgium, October 2018.
42. Mehmet Gunel, Erkut Erdem, and Aykut Erdem, "Language guided fashion image manipulation with feature-wise transformations", *1st Workshop on Computer Vision in Art, Fashion and Design – in conjunction with ECCV 2018*, Munich, Germany, September 2018
43. Hazal Lezki, I. Ahu Ozturk, M. Akif Akpinar, M. Kerim Yucel, K. Berker Logoglu, Aykut Erdem, and Erkut Erdem, "Joint Exploitation of Features and Optical Flow for Real-Time Moving Object Detection on Drones", *2nd International Workshop on Computer Vision for UAVs – in conjunction with ECCV 2018*, Munich, Germany, September 2018.
44. K. Berker Logoglu, Hazal Lezki, M. Kerim Yucel, Ahu Ozturk, Alper Kucukkomurler, Batuhan Karagoz, Erkut Erdem, and Aykut Erdem, "Feature-Based Efficient Moving Object Detection for Low-Altitude Aerial Platforms", *1st International Workshop on Computer Vision for UAVs – in conjunction with ICCV 2017*, Venice, Italy, October 2017.
45. Raffaella Bernardi, Ruket Cakici, Desmond Elliott, Aykut Erdem, Erkut Erdem, Nazli Ikizler-Cinbis, Frank Keller, Adrian Muscat, and Barbara Plank, "Automatic Description Generation from Images: A Survey of Models, Datasets, and Evaluation Measures", *International Joint Conference on Artificial Intelligence (IJCAI 2017)*, Journal track, Melbourne, Australia, August 2017.

46. Mert Kilickaya, Aykut Erdem, Nazli Ikizler-Cinbis, and Erkut Erdem, "Re-evaluating Automatic Metrics for Image Captioning", *The 15th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2017)*, Valencia, Spain, April 2017.
47. Mert Kilickaya, Nazli Ikizler-Cinbis, Erkut Erdem, and Aykut Erdem, "Leveraging Captions in the Wild to Improve Object Detection", *The 5th Workshop on Vision and Language (VL'16) – in conjunction with ACL 2016*, Berlin, Germany, August 2016.
48. Levent Karacan, Aykut Erdem, and Erkut Erdem, "Image Matting with KL-Divergence Based Sparse Sampling", *IEEE International Conference on Computer Vision (ICCV 2015)*, Santiago, Chile, December 2015.
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51. Aysun Kocak, Kemal Cizmeciler, Aykut Erdem, and Erkut Erdem, "Top down saliency estimation via superpixel-based discriminative dictionaries", *British Machine Vision Conference (BMVC 2014)*, Nottingham, UK, September 2014.
52. Yasin Kavak, Erkut Erdem, and Aykut Erdem, "Visual saliency estimation by integrating features using multiple kernel learning", *6th International Symposium on Attention in Cognitive Systems (ISACS 2013)*, Beijing, China, August 2013.
53. Bora Celikkale, Aykut Erdem, and Erkut Erdem, "Visual Attention-driven Spatial Pooling for Image Memorability", *IEEE Computer Vision and Pattern Recognition Workshops (CVPRW 2013)*, Portland, Oregon, USA, June 2013.
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National Conference Publications

1. Tarik Dikkaya, Damla Akcaoglu, Eesen Yuksel, Erkut Erdem, Aykut Erdem, "Ağaç Kanopisi Verilerinde Spektral Rekonstrüksiyonun Performans Analizi: NEON Veri Kümesi Örneği (Performance Analysis of Spectral Reconstruction in Tree Canopy Data: A Case Study on the NEON Dataset)", 34. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2026)*, Istanbul, Turkey, July 2026.
2. Gulse Atun, Abdullah Edik, Emil Cagil Kokdener, Ender Orman, Erkut Erdem, Aykut Erdem, "3B BT Görüntülerinden Türkçe Radyoloji Rapor Üretimi için Çok Kipli Büyük Dil Modelleri (Multimodal Large Language Models for Turkish Radiology Report Generation from 3D CT Images) (in Turkish)", 34. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2026)*, Istanbul, Turkey, July 2026.
3. Idil Aytekin, Onat Dalmaz, Salman Ul Hassan Dar, Aykut Erdem, Erkut Erdem, Tolga Cukur, "Çoklu-Kontrast MRG'de Kanal-Değişim-Ağı ile Görüntü Sentezi (Multi-Contrast MRI Synthesis with Channel-Exchanging-Network)", 30. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2022)*, Safranbolu, Mayıs 2022.
4. Begum Citamak, Menekşe Kuyu, Aykut Erdem, Erkut Erdem, "MSVD-Türkçe: Türkçe Video Altyazılama için Geniş Ölçekli Bir Veri Kümesi (MSVD-Turkish: A Large-Scale Dataset for Video Captioning in Turkish)", 27. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2019)*, Sivas, Nisan 2019.
5. M. Sercan Amac, Aykut Erdem, Erkut Erdem, "Derin Modellerin Eğitiminde Uygulanan Yöntemlerin Giysi Niteliklerinin Tespiti Üzerinde Kıyaslamalı Bir Analizi (A Comparative Analysis of Practices in Training Deep Models for Fashion Attribute Detection)", 27. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2019)*, Sivas, Nisan 2019.
6. Mehmet Günel, Erkut Erdem, Aykut Erdem, "Kişi Görüntülerinin Nitelik Esaslı Üretilmesi (Generating Person Images Based on Attributes)", 26. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2018)*, Çeşme, Mayıs 2018.
7. Ali Yunus Emre Özköse, Tarık Ayberk Yılıkoğlu, Levent Karacan, Aykut Erdem, "Derin Öğrenmeyle Bir Fotoğrafın Yerini Bulma (Finding Location of A Photograph with Deep Learning)", 26. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2018)*, Çeşme, Mayıs 2018.
8. Menekşe Kuyu, Aykut Erdem, Erkut Erdem, "Altsözcük Öğeleri ile Türkçe Görüntü Altyazılama (Image Captioning in Turkish using Subword Units)", 26. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2018)*, Çeşme, Mayıs 2018.
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16. Mert Kılıçkaya, Erkut Erdem, Aykut Erdem, Nazlı İkizler Cinbiş, Ruket Çakıcı, "Meta-sınıf Tabanlı Getirme ile Veriye Dayalı İmge Altyazılama (Data-driven Image Captioning with Meta-class Based Retrieval)", 22. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2014)*, Trabzon, Nisan 2014.
17. Okan Tarhan Tursun, Ahmet Oguz Akyuz, Aykut Erdem, Erkut Erdem, "YDA Görüntüler İçin Gölgeleme Giderme Algoritmalarının Karşılaştırılması (Evaluating Deghosting Algorithms for HDR Images)", 22. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2014)*, Trabzon, Nisan 2014.
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19. Hasan Tuğrul Erdoğan, Erkut Erdem, Aykut Erdem, "Alan Kovaryansları İçin Grup Seyrekliğine Dayalı Seyrek Kodlama (Group Sparsity Based Sparse Coding for Region Covariances)", 21. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2013)*, Girne, KKTC, Nisan 2013.
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22. İ. Aykut Erdem, M. Erkut Erdem, Volkan Atalay, A. Enis Cetin, "Engelliler İçin Bilgisayarlı Görmeye Dayalı Klavye ve Fare Sistemi", 11. *IEEE Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU 2003)*, pp. 196-199, İstanbul, Haziran 2003.

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1. Halit Ozsoy, Mert Cokelek, Inci Ayhan, Erkut Erdem, Aykut Erdem, "How scene attributes and sound influence visual exploration of omnidirectional panoramic scenes", *Annual Meeting of the Cognitive Science Society (CogSci)*, Toronto, Canada, July 2022.
2. Mert Kilickaya, Aykut Erdem, Nazli Ikizler-Cinbis, and Erkut Erdem, "Re-evaluating Automatic Metrics for Image Captioning", *ICCV17 Second Workshop on Closing the Loop Between Vision and Language (CLVL)*, Venice, Italy, October 2017.
3. Cagdas Bak, Aysun Kocak, Aykut Erdem and Erkut Erdem, "Spatio-Temporal Saliency Networks for Dynamic Saliency Prediction", *ICCV International Workshop on 'Mutual Benefits of Cognitive and Computer Vision (MBCC)*, Venice, Italy, October 2017.
4. Sibel Tari, Cagri Aslan, Emre Baseski and Aykut Erdem, "Shape Scale: Representing Shapes at Their Absolute Scales", *SIAM Annual Meeting 2006*, Boston, USA, July 2006.
5. M. Iskar, Erku Erdem, A. Erdem, S. Akman, A. Dickerman, V. Atalay and R. Cetin-Atalay, "CAPRIS: A Database for Cancer Gene Promoter Related Motif Search", *FEBS Journal*, Vol. 273, pp. 248, June 2006.
6. Erkut Erdem, Aykut Erdem and Volkan Atalay, "Image-based Extraction of Material Reflectance Properties of a 3D Rigid Object", *Eurographics 2003*, Poster Session, Granada, Spain, September 2003.
7. Aykut Erdem, Erkut Erdem, Volkan Atalay and A. Enis Cetin, "Computer based text entry for Wearable Computing", *6th IEEE International Symposium on Wearable Computers (ISWC 2002)*, Demonstrations, Seattle, USA, October 2002.
8. Aykut Erdem, Erkut Erdem, Yasemin Yardimci, Volkan Atalay and A. Enis Cetin, "Computer Vision Based Mouse", *IEEE International Conference on Acoustics, Speech, and Signal Processing 2002 (ICASSP 2002)*, Student Forum Session, Florida, USA, May 2002.

Technical Reports

1. Aykut Erdem and Andrea Torsello, "A Game Theoretic Approach To Jointly Learn Shape Categories and Contextual Similarities", Dipartimento di Informatica, Università Ca' Foscari di Venezia, CS-2009-8, July 2009.
2. Aykut Erdem, "A dynamic procedure for forming shape categories from skeletal shape trees", Department of Computer Engineering, Middle East Technical University, METU-CENG-TR-2007-04, July 2007.

Theses

- Aykut Erdem, "Category Knowledge, Skeleton Based Shape Matching and Shape Classification", *PhD Thesis*, Middle East Technical University, Turkey, October 2008.
- Aykut Erdem, "Vision-Based Human-Computer Interaction Using Laser Pointer", *MSc Thesis*, Middle East Technical University, Turkey, July 2003.

Theses Supervised

PhD Theses Supervised

1. Burak Ercan Kocak, "Learning to reconstruct intensity images from events", *PhD Thesis*, (co-supervised with Erkut Erdem),
2. Aysun Kocak, "Dynamic Fusion Networks for Predicting Saliency in Videos", *PhD Thesis*, (co-supervised with Erkut Erdem), Hacettepe University, January 2023.
3. Semih Yagcioglu, "Multimodal Machine Comprehension of How-To Instructions with Images and Text", *PhD Thesis*, Hacettepe University, January 2023.
4. Kemal Cizmeciler, "Leveraging Semantic Saliency Maps for Query-specific Video Summarization", (co-supervised with Erkut Erdem), *PhD Thesis*, Hacettepe University, September 2022.
5. Ismail Bora Celikkale, "Generating Stories from Large Scale Image Collections", *PhD Thesis*, Hacettepe University, February 2020.
6. Yasin Kavak, "Learning Visual Saliency for Static and Dynamic Scenes", *PhD Thesis*, Hacettepe University, September 2017.
7. Osman Akin, "Part-Based Object Tracking with Correlation Filters", *PhD Thesis*, Hacettepe University, August 2016.

MSc Theses Supervised

1. Süleyman Yıldırım, "Generative Augmentation for Blind Image Super-Resolution", MSc Thesis, Koç University, December 2025.
2. Osman Batur Ince, "Dataset Cartography for Compositional Generalization", MSc Thesis, Koç University, July 2024.
3. Halit Ozsoy, "Audio-Visual Saliency in Omnidirectional Panoramic Scenes", (co-supervised with Inci Ayhan), MSc Thesis, Boğaziçi University, May 2024.
4. Burak Can Biner, "Audio-Driven Image Generation and Editing with Pretrained Diffusion Models", MSc Thesis, Koç University, September 2023.
5. Nafiseh Jabbari Tofghi, "Spherical Vision Transformers For Audio-Visual Quality Assessment Of Omnidirectional Image And Video", MSc Thesis, Koç University, September 2023.
6. Mert Cokelek, "Spherical Vision Transformers for Audio-Visual Saliency Prediction in 360° Videos", MSc Thesis, Koç University, September 2023.
7. Ahmed Imam Shah, "Unsupervised Multi-Object Discovery and Tracking using Memory-Augmented Slot Attention", (co-supervised with Yucel Yemez), MSc Thesis, Koç University, September 2023.
8. Abdul Basit Anees, "HyperGAN-CLIP: A Versatile Framework for CLIP-guided Image Synthesis and Editing using Hypernetworks", (co-supervised with Deniz Yuret), MSc Thesis, Koç University, July 2023.
9. Ahmet Canberk Baykal, "GAN Inversion Based Image Manipulation with Text-Guided Encoders", (co-supervised with Deniz Yuret), MSc Thesis, Koç University, August 2022.
10. Burcak Asal, "A Robust Method To Identify Overlapping Crowd Motion Patterns", MSc Thesis, Hacettepe University, January 2017.

11. Cagdas Bak, "Deep Learning Based Saliency Prediction In Videos", *MSc Thesis*, Hacettepe University, July 2016.
12. Mert Kilickaya , "Visual Importance: Modeling With Applications To Vision And Language", (co-supervised with Nazli Ikizler Cinbis), *MSc Thesis*, Hacettepe University, July 2016.
13. Hasan Tugrul Erdogan, "Sparsity-based Discriminative Tracking With Adaptive Cue Integration", (co-supervised with Erkut Erdem), *MSc Thesis*, Hacettepe University, August 2015.
14. Levent Karacan , "Image Smoothing By Using First And Second Order Region Statistics", (co-supervised with Erkut Erdem), *MSc Thesis*, Hacettepe University, January 2014.

Research Grants

As Principal Investigator

■ Developing A Multimodal Large Language Model for Digital Pathology

This project's primary aim is to develop a reliable, interpretable, and interactive multimodal artificial intelligence system capable of simultaneously analyzing pathology images and clinical textual information to improve disease diagnosis accuracy. The proposed system will effectively integrate high-resolution digital pathology images (whole slide images) with clinical narratives such as pathology reports and medical notes, leveraging advanced deep learning approaches. Our model will feature parameter-efficient visual processing techniques capable of analyzing cellular-level details while simultaneously capturing critical contextual information from clinical texts. By jointly processing these modalities, the system will generate clear, interpretable, and clinically relevant outcomes communicated in natural language. This will enable pathologists to rapidly identify essential diagnostic details that might otherwise be overlooked, significantly enhancing diagnostic accuracy and clinical workflow efficiency.

Project Duration: 3 years
(07/11/2025-07/11/2028)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 125E190)

■ VISGEN: Novel Approaches to Visual Data Generation and Editing using Generative Artificial Intelligence

The goal of the VISGEN project is to develop advanced generative AI models for visual data generation and editing by integrating audio inputs, producing high frame-rate long-form videos, and supporting multi-conditional inputs. The project explores applications in image restoration and collaborates with industrial and medical sectors to enhance driving video simulations and clinical diagnostic tools. By combining visual and auditory modalities, the project aims to create more realistic, context-rich synthetic content.

Project Duration: 3 years
(01/06/2025-01/06/2028)

The Scientific and Technological Research Council of Turkey (TUBITAK) The National Outstanding Researchers Program (Award# 123C550)

■ Quality Assessment of 360° Videos Guided by Audio-Visual Saliency

The goal of the project is to investigate automatic evaluation metrics for assessing the quality of the 360° videos experienced through VR devices. Our aim is to contribute to this emerging field of research by exploring the use of human visual attention and multimodal computational saliency models that can process visual

Project Duration: 3 years
(15/03/2021-15/03/2024)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 120E501)

and auditory cues. In this way, the proposed model will make predictions more aligned with human perception. More specifically, we will first develop novel deep neural network architecture for predicting saliency in 360° panoramic videos with ambisonic spatial audio. To train and evaluate our models, we will collect a large video corpus that includes eye fixation data for 360° videos containing 360° spatial audio. This will be yet another contribution to the literature since no such dataset exists and it will facilitate further research on this topic. Secondly, we will collect a new video dataset and develop novel techniques for quality assessment of VR content captured in real environments, which will explicitly exploit the estimated audio-visual saliency maps of the input 360° video.

■ Summarization Approaches Towards Interpreting Big Visual Data

The aim of the proposed project is to interpret big and noisy visual data, which has been recorded in diversified environments with no predefined constraints. To this end, the goal is to develop and apply original data mining methods towards extracting important knowledge and increase the accessibility of such archives. Particularly, we aim to focus on summarization approaches, so that the big visual data is more effectively structured and enriched with additional semantic information. The summarization approaches that make use of the multi-modal nature of the data will focus on three main problems: 1) To learn semantic concepts and spatio-temporal attributes from big visual data; 2) organization of large photograph collections; 3) summarization of videos in large web archives. In all these problems, big visual data and the additional information referred as metadata will be handled together.

Project Duration: 3 years
(15/09/2017-15/09/2020)

The Scientific and Technological Research Council of Turkey (TUBITAK) Primary Subjects R&D Funding Program (Award# 116E685)

■ Towards A Unified Framework For Finding What Is Interesting In Videos

The goal of this project is to develop and apply effective computer vision techniques that can automatically detect "what is interesting" in such videos. Here, what is meant by interesting may refer to different notions such as where people look in videos, salient objects, interesting motion patterns or moments in videos. All these topics listed above form the subject matter of the proposed project. It is important to note that each notion of interestingness involves different computational problems that need to be solved. This project, unlike the previous work, will investigate all these interrelated concepts within a unified framework and this will allow us to detect different levels of interestingness in a more accurate way. Although the aforementioned problems are closely related to each other, most of the existing literature treats them separately – as mentioned above. In this project, we bridge this gap by developing various techniques and methodologies for solving each task, which take advantage of simultaneously using additional information from other sources of interestingness.

Project Duration: 3 years
(01/04/2014-01/04/2017)

The Scientific and Technological Research Council of Turkey (TUBITAK) Career Development Program (Award# 113E497)

As Co-Investigator

■ End-to-End Approaches for Hyperspectral Image Enhancement and Synthesis

The aim of the project is to develop novel end-to-end approaches for hyperspectral image enhancement. Furthermore, we will also explore the exciting possibility of synthesizing hyperspectral images from standard RGB images, which has received limited attention in the literature, but has the potential to greatly expand the capabilities of hyperspectral imaging. To achieve our goals, we will leverage recent advances in deep learning and computer vision, including vision transformers, self-supervised learning, and diffusion-based generative modeling.

Project Duration: 3 years
(01/10/2023-01/10/2026)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 123E385)

■ Seeing Through Events: End-to-End Approaches to Event-Based Vision Under Extremely Low-Light Conditions

The aim of the project is to provide novel ways to process event data using deep neural networks and will offer hybrid approaches to bring traditional cameras and event cameras together to solve crucial challenges we face when capturing and processing videos in dark. The neural architectures that will be explored in this research project can also be applied to other event-based computer vision tasks.

Project Duration: 3 years
(15/03/2022-15/03/2025)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 121E454)

■ Using Synthetic Data for Deep Person Re-Identification

The aim of the proposed project is to produce new large-scale datasets which will allow construction of more powerful deep models for person re-identification and are much larger and more comprehensive than the existing ones. For this purpose, instead of manually labeling images taken from cameras placed at different locations under specific scenarios, in our project, we will investigate novel synthetic data generation methods.

Project Duration: 2 years
(01/06/2018-01/06/2020)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 217E029)

■ A Multimodal and Multilingual Framework for Video Captioning

In this project, we claim that novel automatic description generation approaches should be developed for low-resource, highly inflected and highly agglutinative languages to further boost research on integrated vision and language. We aim to contribute to this area of research by exploring video captioning approaches, with a special focus on the Turkish language. With this objective, first, we will develop novel video captioning approaches, which can deal with language-specific properties of Turkish. Secondly, we will study cross-lingual video captioning where the models exploit descriptions in English as an additional source of information during the caption generation process.

Project Duration: 2 years
(01/06/2018-01/06/2020)

The Scientific and Technological Research Council of Turkey (TUBITAK) and British Council - Newton-Katip Çelebi Fund Institutional Links Grant Programme (Award# 217E054)

■ Understanding Images and Visualizing Text: Semantic Inference and Retrieval by Integrating Computer Vision and Natural Language Processing

This project will explore the connection between vision and language from different directions in which we will integrate computer vision and natural language processing methods. By using these two fields together, automatic systems that transcribe the visual content of images with vivid descriptions which are very alike to human language will be obtained. Similarly, in the context of this project,

Project Duration: 3 years
(01/10/2013-01/10/2016)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 113E116) and European Union under European Cooperation in Science and Technology (COST) Programme (ICT COST IC1037 Action)

retrieval systems that describe the sentence or paragraph-based textual queries visually via related images or image sets will be constructed.

■ The Use of Multiple Cues and Contextual Knowledge in Computer Vision

This project will explore the influences of visual context and multiple cues on a number of computer vision problems. First, a novel visual saliency or attention model will be developed towards a direction that combines information coming from multiple cues with the contextual knowledge. The goal is to come up with a model that can effectively predict where people look in an image. In the second part of the project, we will investigate the problem of image filtering with a focus on devising appropriate ways of extracting high-level contextual knowledge for filtering and using them to guide the ongoing image smoothing process. The third part of the project will be about developing adaptive approaches to image segmentation that integrates information obtained from multi cues. A novel and effective segmentation algorithm will be developed that adaptively combines high-level prior knowledge with the information obtained from different visual cues at different scales.

Project Duration: 3 years
(01/10/2012-01/10/2015)

The Scientific and Technological Research Council of Turkey (TUBITAK) Career Development Program (Award# 112E146)

As Consultant

■ Design and Experimental Demonstration of Phase Balanced Optical Power Splitters for Next-Generation Nanophotonic Technologies Through Machine Learning

The aim of the proposed project is to develop a new photonic simulation method that is at least 100 times faster than 3D simulations but with matching physical accuracy. This method will be realized using a data driven deep learning model, and will create a paradigm shift in nanophotonics and electromagnetic design.

Project Duration: 3 years
(31/03/2022-31/03/2025)

The Scientific and Technological Research Council of Turkey (TUBITAK) The Support Program for Scientific and Technological Research Projects (Award# 122E214)

Professional Activities

Editorial Work for Journals

- Senior Area Editor, , IEEE Transactions on Image Processing (2025-now)
- Associate Editor, IEEE Transactions on Image Processing (2022-2025)

Reviewer for Journals

- Elsevier Computers & Graphics
- Elsevier Computer Vision and Image Understanding
- Elsevier Pattern Recognition
- IEEE Computer Graphics and Applications
- IEEE Robotics and Automation Letters
- IEEE Transactions on Image Processing

- IEEE Transactions on Multimedia
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- IET Computer Vision
- International Journal of Pattern Recognition and Artificial Intelligence Pattern Recognition
- Signal Processing: Image Communication

Program Committee

- 6th Workshop on Vision and Language (VL'17)
- 5th Workshop on Vision and Language (VL'16)
- 3rd International Workshop on Similarity-Based Pattern Analysis and Recognition (SIMBAD 2015)
- 2nd International Workshop on Similarity-Based Pattern Analysis and Recognition (SIMBAD 2013)

Area Chair for Conferences

- Conference on Language Modeling (COLM), 2025
- IEEE International Conference on Image Processing (ICIP), 2024

Reviewer for Conferences

- Annual Conference of the European Association for Computer Graphics (Eurographics), 2025
- Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 2021
- Annual Conference on Neural Information Processing Systems (NeurIPS) 2025
- Annual Meeting of the Association for Computational Linguistics (ACL) 2019, 2020, 2021, 2024
- Asian Conference on Computer Vision (ACCV), 2016
- Asian Conference on Machine Learning (ACML), 2020, 2022
- British Machine Vision Conference (BMVC), 2018, 2017, 2022
- Conference on Empirical Methods in Natural Language Processing (EMNLP) 2019, 2020, 2021, 2022
- Conference on Language Modeling (COLM) 2024
- Conference of the European Chapter of the Association for Computational Linguistics (EACL) 2021, 2024
- European Conference on Computer Vision (ECCV), 2018, 2016
- IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, 2017, 2016, 2015, 2014
- IEEE International Conference on Computer Vision (ICCV), 2025, 2017, 2015, 2009
- IEEE International Conference on Robotics and Automation (ICRA), 2020, 2017, 2013
- IEEE Winter Conference on Applications of Computer Vision (WACV), 2016, 2015, 2014, 2013
- International Conference on Pattern Recognition (ICPR), 2016, 2010
- International Conference on Learning Representations (ICLR), 2026
- SIGGRAPH, 2023
- SIGGRAPH Asia, 2018, 2022, 2025
- SIGNLL Conference on Computational Natural Language Learning (CoNLL), 2019, 2020

Memberships

- 2015 – present The Association for Computational Linguistics
- 2013 – present The Computer Vision Foundation
- 2022 – present IEEE Signal Processing Society